

Mercedes F1 M12 E Performance Analysis

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August 9, 2021

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1 Executive Summary

The aim of this paper is to take a thermodynamic subject and break it down enough to verify whether the claims made in the article being explored are accurate. Specifications and a few operation characteristics are also given, which will make the initialization of the thermodynamic analysis easier for us to begin without the need to make too many assumptions. This is an important topic to analyze because hundreds of millions of dollars are invested annually by different manufacturers to create the most efficient machines possible so we should be able to see what kind of technicality and expertise the R&D teams are working with [3]. In this paper I will go through the analysis at various state points on the internal combustion engine (ICE) and verify any claims being made across the media by either Mercedes or bloggers.

2 Background on Formula 1 Performance

Formula 1 engines are known by car enthusiasts for their high pitched exhaust notes down the straight strips of a circuit and the wonderful downshift exhaust notes around corners. From the year 1947 through the present times the goal of Formula 1 has been to create the most powerful and efficient engines possible which would take a lightweight vehicle across the finish line of a race track. Tons of development was made by all the well-known manufacturers (Mercedes-Benz, Ferrari, Ford, Honda) of which they all take continued to this day to make improvements not just in the motorsport but on the public streets as well. Engines of all different displacements were allowed up until 2014 where the Fédération Internationale de l'Automobile (FIA) will only allow 1.6L 4-stroke piston V6 engines to compete[5]. The reasoning behind this decision was to mitigate emissions and make fuel combustion more efficient so the global warming circumstance lessens moving into the future.

The most interesting part about this move is that it would put Formula 1 at the pinnacle of thermodynamic waste heat recovery system (WHRS) breakthroughs and to really push the boundaries in efficiencies for a 1.6L 4-stroke V6 engine - from which all engineers in the automotive can learn from and apply concepts to road vehicles. The power unit (PU) which consists of the internal combustion engine (ICE), turbocharger (TC), motor-generator unit-heat (MGU-H), motor-generator unit-kinetic (MGU-K), energy store (ES) and controls electronics (CE). Figure 1 shows this PU for the Mercedes M12 E which we will be analyzing later.

2.1 Performance Claims

Mercedes has claimed to have created the first ever 50% thermally efficient power unit in Formula 1[4]. What does this mean? Half of whatever fuel source is used (high octane unleaded in the case of F1) is used

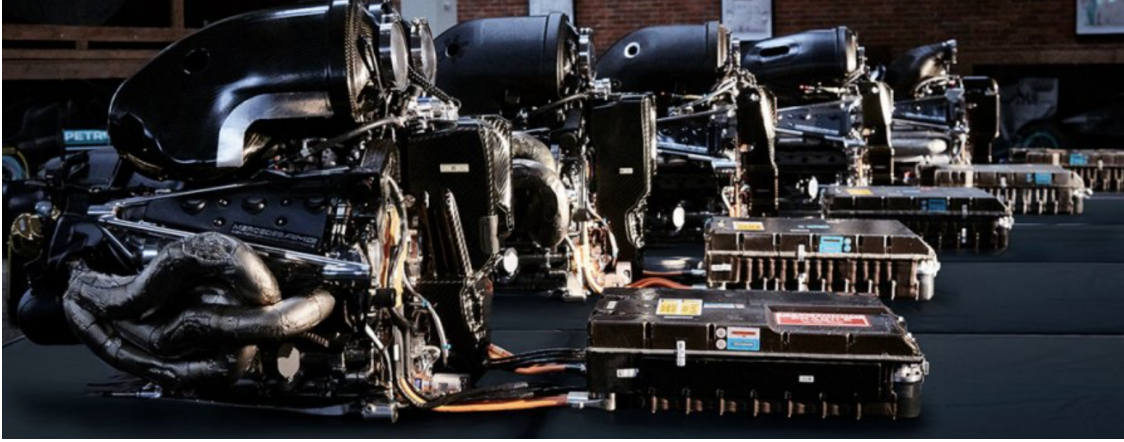


Figure 1: Mercedes M12 E PU

as useful energy and the other half is wasted. For comparison, take a regular street car which may be in the range of 20-35% efficiency - these cars only utilize 20-35% of the energy source as useful energy while the rest is wasted as heat. One of the biggest challenges for present engineers is figuring out ways in which to reduce friction and other sources of heat loss in engines to increase the thermal efficiency from the 20-35% range. The Mercedes F1 M12 E engine under evaluation in this paper has 50% thermal efficiency when looking at the PU as a whole. Without the MGU-H and MGU-K, what would the ICE thermal efficiency be? These are the ideas we will be exploring today. In addition, we will include some analysis on the WHRS and how much energy it reintroduces to the system.

3 Analysis of the Mercedes F1 M12 E

3.1 Specifications of Mercedes F1 M12 E PU

Figure 2 shows the specifications for the ICE taken from Mercedes directly[2]. According to Mercedes, the temperatures get as hot as 2,600C, or approximately 5,200R, at the time of combustion which will be our T3 for our spreadsheet input. The compressor is also able to reach pressures up to $5\times$ atmospheric pressure[1]. We are looking to aim for an Otto thermal efficiency in the 45% ballpark but definitely no more than 48%. We also are looking for engine power to be around 600 horsepower (HP) which might be quite conservative considering some legal road cars like Lamborghini's are stock with 600+ HP. With the turbocharger at peak performance we would like to be in the 850 - 1000 HP range as per specifications listed in Wikipedia[5].

3.2 Thermodynamic Analysis of Mercedes F1 M12 E ICE and Turbocharger

The F1 ICE discussed in an Otto engine, which means our cutoff ratio must be equal to 1 since the specific volume v_4 equals v_3 . After some research the compression ratio commonly used in F1 engines is in the 12:1 - 13:1 range and for this analysis we use $V_r = 12$. Our specific heats at constant pressure and constant volume are 0.24 and 0.17 respectively which means our specific heat ratio, k , equals 1.4. The engine sizing, speed and displacement is taken directly from Mercedes[2]. The fuel heat content of high octane unleaded was assumed to be used. The air-to-fuel ratio used is 14, and given that FIA has restricted the engines to 100kg/hr of fuel, we can solve for the mass air rate.

The first state in the Otto cycle was assumed to be close to atmospheric temperature and atmospheric pressure because it is at the largest specific volume in the cycle and the starting point of the adiabatic compression of the cycle - hence we assume $p_1 = 60$ psia and $T_1 = 1100$ R. Solving for specific volume v_1 we get:

$$v_1 = \frac{R_u \times T_1}{p_1 \times N \times 144 \frac{in^3}{ft^3}} = \frac{1545 \frac{ft \cdot lb}{lb \cdot mol \cdot R} \times 1100 R}{60 \frac{lb}{in^3} \times 28.966 \frac{g}{mol} \times 144 \frac{in^3}{ft^3}} = 6.791 \frac{ft^3}{lbm} \quad (1)$$

Due to the compression ratio of 12 we can find v_2 by the following:

$$v_2 = \frac{v_1}{12} = 0.566 \frac{ft^3}{lbm} \quad (2)$$

Now knowing v_1 , v_2 and p_1 we can find p_2 :

$$p_2 = p_1 \left(\frac{v_1}{v_2} \right)^k = 6.791 \frac{ft^3}{lbm} \left(\frac{6.791}{0.566} \right)^{1.41} = 1,944 \text{ psia} \quad (3)$$

Temperature T_2 is calculated such as:

$$T_2 = \frac{p_2 \times v_2 \times N \times 144 \frac{in^3}{ft^3}}{T_1} = \frac{1,944 \text{ psia} \times 6.791 \frac{ft^3}{lbm} \times 28.966 \frac{g}{mol} \times 144 \frac{in^3}{ft^3}}{1,100 R} = 4,279.5 \text{ R} \quad (4)$$

In order to find what p_3 equals we need to use our best judgement to judgement and choose a T_3 that is appropriate considering that state 3 has already gone through adiabatic expansion and fuel has been injected and combusted. Mercedes has mentioned that the instantaneous gas temperatures at the time of combustion reach 2,600C which is approximately 5,200R[1]. However, it is quite possible that they could get even hotter and for this analysis we will assume the temperature reaches approximately 4,700C. Now we can find p_3 by:

$$p_3 = \frac{R_u \times T_3}{v_3 \times N \times 144 \frac{in^3}{ft^3}} = \frac{1545 \frac{ft \cdot lb}{lb \cdot mol \cdot R} \times 9,000 \text{ R}}{0.566 \frac{ft^3}{lbm} \times 28.966 \frac{g}{mol} \times 144 \frac{in^3}{ft^3}} = 5,890.9 \text{ psia} \quad (5)$$

On our excel spreadsheet containing the results we see a fourth state-this state is collapsed into state 3 when choosing a expansion ratio of 1 to convert the model from a dual pressure diesel cycle to an Otto cycle, hence the values remain the same at state 4. The final state which brings us full circle is state 5, where $v_5 = v_1$. p_5 is calculated using equation (3) but substituting v_1 and v_2 for v_3 and v_5 and p_1 for p_3 -this tells us $p_5 = 117.2$ psia. Finally to complete the process table we find T_5 using equation (4) but substituting p_2 and v_2 for p_5 and v_5 and T_1 for T_4 . $T_5 = 3,249$ R. Figure xy shows the results for the given assumptions above and the resulting property table, process table, turbocharger calculations, Otto cycle efficiency and engine power.

The calculations for the process table were all the same. Find the internal energy was nothing more than the following equation:

$$u_{i_1 i_2} = C_v \times (T_{i_2} - T_{i_1}) \quad (6)$$

where i_2 represents the state proceeding a state i_1 . Work is only done from states 1-2 and 3-5 because of the adiabatic process which can be represented by:

$$W[BTU] = \frac{((p_2 \times v_2) - (p_1 \times v_1)) \times 144 \frac{in^3}{ft^3}}{(1 - k) \times 778 \frac{ft \cdot lb}{BTU}} \quad (7)$$

Now heat can be represented as the sum of internal energy and work. With the property and process tables now complete we can calculate the Otto cycle efficiency η_{otto} and engine power. The Otto cycle efficiency is expressed by the following equation:

$$\eta_{otto} = 1 - \frac{Q_{in} + Q_{out}}{Q_{in}} = 1 - \frac{Q_{2,3} + Q_{5,1}}{Q_{2,3}} \quad (8)$$

The engine power is represented as

$$W[HP] = \frac{Q_{23} \times D_{engine} \times Vol_{eff} \times 2 \times RPM \times 60 \times Effw.Losses}{v_1 \times Number_{strokes} \times 2545 \times 1728} \quad (9)$$

We can represent the Torque T by

$$T = \frac{W[HP] \times 5252}{RPM} \quad (10)$$

Lastly is the turbocharger, which drives the power up and also has its set of assumptions. It will be operating in ambient conditions with 85% efficiency and 82% turbo expander efficiency. Knowing the exhaust temperature $T_{exhaust}$ is critical to help us see if our assumptions are good ones in that they are close to what the real world values are. To find $T_{exhaust}$ we need to find the change in enthalpy ΔH by the following

equation:

$$\Delta H = C_p \times T_5 \times \left(1 - \left(\frac{14.7psia}{p_5}\right)^{\frac{k-1}{k}}\right) \times TurboExpanderEff \quad (11)$$

Now we can calculate $T_{exhaust}$ by:

$$T_{exhaust} = T_5 - \frac{\Delta H}{C_p} \quad (12)$$

The most important parameter we need from the turbocharger are p_{out} and T_{out} which represent the dynamically changing variables that become the new input to p_1 and T_1 in the property table. To calculate p_{out} and T_{out} we use:

$$p_{out} = 14.7psia \times \left(1 + \frac{(1 - TurboBypass) \times \Delta H \times CompressorEff}{C_p \times 540R}\right)^{\frac{k}{k-1}} \quad (13)$$

$$T_{out} = 540R + \frac{540R \times \left(\left(\frac{p_{out}}{14.7psia}\right)^{\frac{k-1}{k}} - 1\right)}{CompressorEff} \quad (14)$$

4 Discussion of Results

4.1 Assumptions Made

Assumptions had to be made in order to model the Mercedes Formula 1 M12 E and some research was done in order to figure out appropriate variables, such as an approximation for T_1 and P_1 , temperature which the ICE reaches at peak performance, compression ratio, exhaust temperature, air-fuel ratio, turbo expander efficiency, turbo bypass percentage and compressor efficiency.

The initial assumption that $p_1 = 60$ psia seems a bit low but it could be possible because at state 1 the piston is at the largest specific volume and has just finished its adiabatic expansion. The temperature T_1 seems reasonable for the chosen p_1 because it is the coolest temperature in the entire cycle. These two assumptions also make sense because the work done from states 1 to 2 is negative, meaning some energy is required to complete the compression from state 1 to 2, and at the same time there is a significant amount of internal energy created from that adiabatic compression.

The assumption for T_3 is perhaps the most worrisome assumption that there is in this whole model because it represents 4,600C. The sun is approximately 5,600C but the only reason why I have stuck to this value for T_3 is because it gives us the exact answers to Otto cycle efficiency, engine power and exhaust temperature which we are looking for. And although 4,600C is ridiculously hot, there are many cooling systems onboard the F1 vehicles - from water cooling to massive air flow rates. 4,600C is also not a value which will persistently show up at every combustion in the engine; this is assuming full throttle, 15,000 RPM

on a straightaway of a circuit.

The pressure p_3 comes out to about 5,900 psia which is very high. However, this can be considered possible because according to Mercedes the turbocharger is designed to induce up to $5\times$ atmospheric pressure[1]. When we begin with more pressure at the initial state 1 we produce much more pressure and heat in states 2 and 3 and releases less heat as well as shown in the process tables in Figures 5-9 because more energy can combust. If we take a look at the progressive change in the Otto efficiency (cell B11), horsepower (c30), torque (c31) and $T_{exhaust}$ (k36) from Figures 5-9 we notice the efficiency increases from .45 to .47, the horsepower produced increases from 584 to 1,056 hp, torque increases from 204 to 370 ft-lbf and the $T_{exhaust}$ is in the range of 1366-1150 Rankine.

4.2 Do The Results Make Sense?

If we take the Wikipedia site with the most up to date F1 specifications[5] set by FIA, we see that power output of the ICE is in the 875-1000 hp range. The set of input used in Figure 5 where we were not using turbocharger input for the property table shows engine power to be 584 hp. This is very likely on the lower end of what the F1 engines are producing naturally because there are 3.8L V6 engines like that of a Nissan GT-R which produce this horsepower out of the dealership. However, once we get as close as we can to steady state with the turbocharger input to the property table (as shown from the three iterations given from Figures 6-9) we see a very large increase in hp and even close to 1,100 hp. This nearly 500 hp addition of horsepower from the turbocharger is quite high. If we brought down the turbocharger efficiency enough to get us in the 200hp range then these numbers would be much more appealing and would put us at nearly 800 hp with the turbocharger active.

If we evaluate the Otto cycle efficiency we see are in the range of 45%-47%. This is less than the claimed 50% thermal efficiency that this entire project is made to prove or disprove. However, the claim of 50% thermal efficiency also includes the MGU-H and MGU-K which do reintroduce a good portion energy into the system. Wikipedia notably mentions that the maximum retrievable amount of energy from the MGU-K is 120 MW, which if taken into account for in our Otto cycle efficiency equation it is certainly possible for the thermal efficiency to reach 50%! The torque typically seen from these F1 engines is around the 440-480 ft-lb range, and as seen from Figure 9 we are around that ballpark. In addition, the $T_{exhaust}$ temperatures are in the 1150-1350 Rankine range. Those temperatures correspond to a blue and/or dark blue flame that comes out of the exhaust of the F1 engines and those temperatures are where those flames are seen which is also another indicator that the assumptions and conclusions we have made are in the ballpark of what is really going on internally in these engines.

5 Conclusion

In conclusion, Mercedes' claim to to have achieved 50% thermal efficiency is highly likely. This model is not accurate at all however it has given us one solution which has led us to all the direct specifications that Mercedes and FIA has given the public. The most significant sources of error in this model come from the value of T_3 and the efficiencies assumed for the turbocharger. Without the turbocharger and with some more modification to the assumptions made in the property table there should be a fair shot at modeling this engine as close to the real thing as possible. The turbo efficiencies are also not accurate at all as we were not given them. The big picture is spectacular when in retrospect, who would have ever thought 50% thermal efficiency would be possible? Engine cooling causes a lot of useful energy to be wasted but if there is some kind of absorbent technology that is innovated in the future which could somehow convert that energy into stored energy then breaking the 50% benchmark would be possible. This is not only useful for the motorsport world, but if these technologies can be made cheap and readily available then perhaps regular everyday driving could be more efficient and keep petroleum vehicles in the market much longer than what is presently thought when considering EVs and climate change.

6 Appendix

Internal Combustion Engine (ICE)

Capacity:	1.6 litres
Cylinders:	Six
Bank Angle:	90
No of Valves:	24
Max rpm ICE:	15,000 rpm
Max Fuel Flow Rate:	100 kg/hour (above 10,500 rpm)
Fuel Injection:	High-pressure direct injection (max 500 bar, one injector/cylinder)
Pressure Charging:	Single-stage compressor and exhaust turbine on a common shaft
Max rpm Exhaust Turbine:	125,000 rpm

Figure 2: Mercedes M12 E ICE Specifications

Combustion, construction, operation, power, fuel and lubrication [\[edit \]](#)

- **Manufacturers:** [Mercedes](#), [Renault](#), [Ferrari](#) and [Honda](#)
- **Type:** Hybrid-powered intercooled
- **Engine stroke combustion:** Four-stroke piston Otto cycle
- **Configuration:** [V6](#) single [hybrid turbocharger](#) engine
- **V-angle:** 90° cylinder angle
- **Displacement:** 1.6 L (98 cu in)
- **Bore:** Maximum 80 mm (3.15 in)
- **Stroke:** 53 mm (2.09 in)
- **Valvetrain:** [DOHC](#), 24-valve (four valves per cylinder)
- **Fuel:** 98–102 RON unleaded gasoline + 5.75% biofuel
- **Fuel delivery:** [Gasoline direct injection](#)
- **Fuel injection pressure:** 500 bar (7,252 psi; 493 atm; 375,031 Torr; 50,000 kPa; 14,765 inHg)
- **Fuel-mass flow restrictor rate:** 100 kg/h (220 lb/h) (−40%)
- **Fuel economy mileage range:** 6 mpg-US (39.20 L/100 km)
- **Aspiration:** Single-turbocharged
- **Power output:** 875–1,000 + 160 hp (652–746 + 119 kW) @ 10,500 rpm
- **Torque:** Approx. 600–650 N·m (443–479 lb·ft)
- **Lubrication:** [Dry sump](#)
- **Maximum revs:** 15,000 RPM
- **Engine management:** McLaren TAG-320 (2014-2018) later TAG-320B (2019-present)
- **Max. speed:** 370 km/h (230 mph) (Monza, Baku and Mexico); 340 km/h (211 mph) normal tracks
- **Cooling:** Single mechanical water pump feeding a single-front cooling system
- **Ignition:** High energy inductive
- **Banned engine materials:** Magnesium-based alloys, Metal Matrix Composites (MMCs), intermetallic materials, alloys containing more than 5% by weight of platinum, ruthenium, iridium or rhenium, copper-based alloys containing more than 2.75% beryllium, any other alloy class containing more than 0.25% beryllium, tungsten-based alloys and ceramics, and ceramic-matrix composites
- **Weight:** 145 kg (320 lb) overall including headers, clutch, ECU, spark box or filters

Figure 3: General F1 Engine Specifications By FIA

Forced induction and push-to-pass [\[edit \]](#)

- **Turbocharger vendors:** [Garrett Motion \(Ferrari\)](#), [IHI Corporation \(Honda\)](#), [Mercedes AMG HPP \(in-house Mercedes\)](#) and [Pankl Turbosystems GmbH \(Renault\)](#)
- **Turbocharger weight:** **8 kg (18 lb)** depending on the turbine housing used
- **Turbocharger spin rev limit:** 125,000 rpm
- **Pressure charging:** Single-stage compressor and exhaust turbine, a common shaft
- **Turbo boost level pressure:** Unlimited but mainly typical 4.0 to 5.0 **bar** (58.02 to 72.52 **psi**; 3.95 to 4.93 **atm**; 3,000.25 to 3,750.31 **Torr**; 400.00 to 500.00 **kPa**; 118.12 to 147.65 **inHg**) absolute
- **Wastegate:** Maximum of two, electronic or pneumatic controlled

Figure 4: General F1 Turbo Specifications By FIA

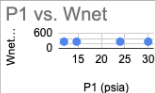
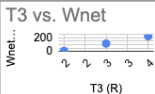
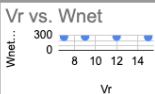
	A	B	C	D	E	F	G	H	I	J	K	L	
1	Vr	12.000		NOTE: ANYTHING IN BOLD IS INPUT. ALL ELSE DEPENDS ON INPUT IF k == comprss index or expansionindex, there is no heat transfer Internal eenergy = Cv(deltaT)					Vr compression ratio v1/v2				
2	Vc	1.000							Vc cutoff ratio v4/v3				
3	k	1.412							Cp specific hear constant p				
4	Cp	0.240							Cv specific heat constant v				
5													
6	Cv	0.170		PROPERTY TABLE					k specific heat ratio Cp/Cv				
7	Mole Wt.	28.966											
8	Ru	1545.000			p(psia)	T(R)	v(ft³/Lbm)						
9	Compress. Index	1.410	n, 1-2 nc	1.000	60.000	1100.000	6.791		(n-1)/n, expansion	0.2907801418			
10	Expansion Index	1.410	n, 4-5 ne	2.000	1994.329	4279.498	0.566						
11	otto Cycle Eff.	0.455		3.000	5890.909	9000.000	0.566		P1				
12	Cycle Efficiency	0.625		4.000	5890.909	9000.000	0.566						
13	Efficiency	0.793		5.000	177.230	3249.213	6.791						
14	Eng. Eff Corr		<--WorkOUT/HeatIN						T3				
15	Carnot Effic	0.878		PROCESS TABLE						Vr			
16	Exergy in Exhaust			Heat	Internal Energy	Work							
17	Eff w losses	0.450		1-2	214.964	540.515	-325.550						
18	BSFC(LBm/hr/hp)			2-3	802.485	802.485	0.000						
19	Intercooler effectiveness			3-4	0.000	0.000	0.000						
20	Closed mass energy			4-5	-16.013	-977.634	961.621						
21				5-1	-365.366	-365.366	0.000						
22													
23				SUM	636.070	0.000	636.070						
24													
25				ENGINE SIZING					Turbo Activated ? no				
26	Volumetric eff	1.100			4.000	No. Cycles			TURBOCHARGER STUFF				
27	2 stroke or 4 stroke engine?	4.000			3.150	Bore			Tmin Exh. Cooling [R]				
28	Engine speed(rpm)	15000.000			2.087	Stroke			Ambient Temp 540				
29	Engine disp. (in³)	97.600							Ambient Pressure 14.7				
30	Engine power(hp, kW)	584.190	435.638			No. of Engines	1.000		Cycle cooling coeff. 0.82				
31	Engine torque(Ft-Lbf)	204.544				Radiator DT	25.000		Turbo Expander Efficiency 0.85				
32	Brake Mean Eff Pressure (psia)					Twater,in	175.000		Turbo Bypass "F" 0.71				
33	Piston speed(ft/min,m/s)					Twater,out			Compressor Efficiency 0.85				
34	Heat content(btu/ft³)	899158.000				PUMP EFF	0.850		Cycle Mix Temp (T1) R 676.6677602				
35	Air/Fuel ratio	14.000				Rad. Ht Rec. Massflow			delta H 341.4651153				
36	Mair	3686.44	LBm/Hr			Dp=cond	10.000		Texhaust Out (R, F) 1826.442 1366.442				
37	Mfuel	220.46	LBm/Hr			hp	750.000		Tmix (R, F) 2836.609442 2376.609442				
38													
39						Closed Mass Energy			Pout (Psia) 82.18053701				
40						Seconds/Cycle			@Tout (R, F) 952.603681 492.603681				
						BSFC (LBm/Hr/Hp)							

Figure 5: Regular Engine Operation WITHOUT Turbocharger Activated

	A	B	C	D	E	F	G	H	I	J	K	L
1	Vr	12.000		NOTE: ANYTHING IN BOLD IS INPUT. ALL ELSE DEPENDS ON INPUT IF k = comprss index or expansionindex, there is no heat transfer Internal eergy = Cv(deltaT)						Vr compression ratio v1/v2		
2	Vc	1.000								Vc cutoff ratio v4/v3		
3	k	1.412								Cp specific hear constant p		
4	Cp	0.240								Cv specific heat constant v		
5												
6	Cv	0.170		PROPERTY TABLE						k specific heat ratio Cp/Cv		
7	Mole Wt.	28.966			p(psia)	T(R)	v(ft³/Lbm)					
8	Ru	1545.000		1.000	82.200	952.600	4.293		(n-1)/n, expansion	0.2907801418		
9	Compress. Index	1.410	n, 1-2 nc	2.000	2732.231	4279.498	0.358					
10	Expansion Index	1.410	n, 4-5 ne	3.000	9319.337	9000.000	0.358					
11	otto Cycle Eff.	0.487		4.000	9319.337	9000.000	0.358					
12	Cycle Efficiency	0.626		5.000	280.375	3249.213	4.293					
13	Efficiency	0.847										
14	Eng. Eff Corr		<--WorkOUT/HeatIN									
15	Carnot Effic	0.894		PROCESS TABLE								
16	Exergy in Exhaust			Heat	Internal Energy	Work						
17	Eff w losses	0.450		1-2	283.646	565.573	-281.927					
18	BSFC(LBm/hr/hp)			2-3	802.485	802.485	0.000					
19	Intercooler effectiveness			3-4	0.000	0.000	0.000					
20	Closed mass energy			4-5	-16.013	-977.634	961.621					
21				5-1	-390.424	-390.424	0.000					
22												
23				SUM	679.694	0.000	679.694					
24												
25				ENGINE SIZING								
26	Volumentic eff	1.100			4.000	No. Cycles			Turbo Activated ?	no		
27	2 stroke or 4 stroke engine?	4.000			3.150	Bore			TURBOCHARGER STUFF			
28	Engine speed(rpm)	15000.000			2.087	Stroke			Tmin Exh. Cooling [R]			
29	Engine disp. (in³)	97.600			No. of Engines	1.000			Ambient Temp	540		
30	Engine power(hp, kW)	924.181	689.173		Radiator DT	25.000			Ambient Pressure	14.7		
31	Engine torque(Ft-Lbf)	323.586			Twater,in	175.000			Cycle cooling coeff.	0.82		
32	Brake Mean Eff Pressure (psia)				Twater,out				Turbo Expander Efficiency	0.85		
33	Piston speed(ft/min,m/s)				PUMP EFF	0.850			Turbo Bypass "F"	0.71		
34	Heat content(btu/ft³)	899158.000			Rad. Ht Rec. Massflow				Compressor Efficiency	0.85		
35	Air/Fuel ratio	14.000			Dp=cond	10.000			Cycle Mix Temp (T1) R	676.6677602		
36	Mair	3086.44	LBm/Hr		hp	750.000			delta H	381.5930299		
37	Mfuel	220.46	LBm/Hr						Texhaust Out (R, F)	1659.242	1199.242	
38									Tmix (R, F)	2788.121545	2328.121545	
39					Closed Mass Energy				Pout (Psia)	96.01225231		
40					Seconds/Cycle				@Tout (R, F)	1001.091578	541.0915778	
					BSFC (Lbm/Hr/Hp)							

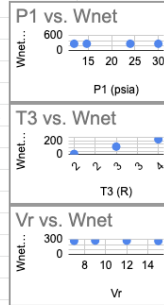


Figure 6: Regular Engine Operation WITH Turbocharger Activated

	A	B	C	D	E	F	G	H	I	J	K	L	
1	Vr	12.000		NOTE: ANYTHING IN BOLD IS INPUT. ALL ELSE DEPENDS ON INPUT If k == comprss index or expansionindex, there is no heat transfer Internal energy = Cv(deltaT)						Vr compression ratio v1/v2			
2	Vc	1.000								Vc cutoff ratio v4/v3			
3	k	1.412								Cp specific hear constant p			
4	Cp	0.240								Cv specific heat constant v			
5													
6	Cv	0.170		PROPERTY TABLE						k specific heat ratio Cp/Cv			
7	Mole Wt.	28.966			p(psia)	T(R)	v(ft^3/Lbm)						
8	Ru	1545.000		1.000	96.000	1001.100	3.863		(n-1)/n, expansion	0.2907801418			
9	Compress. Index	1.410	n, 1-2 nc	2.000	3190.926	4279.498	0.322						
10	Expansion Index	1.410	n, 4-5 ne	3.000	10356.608	9000.000	0.322						
11	otto Cycle Eff.	0.476		4.000	10356.608	9000.000	0.322						
12	Cycle Efficiency	0.626		5.000	311.582	3249.213	3.863						
13	Efficiency	0.829											
14	Eng. Eff Corr		←WorkOUT/HeatIN										
15	Carnot Effic	0.889		PROCESS TABLE									
16	Exergy in Exhaust				Heat	Internal Energy	Work						
17	Eff w losses	0.450		1-2	261.047	557.328	-296.280						
18	BSFC(LBm/hr/hp)			2-3	802.485	802.485	0.000						
19	Intercooler effectiveness			3-4	0.000	0.000	0.000						
20	Closed mass energy			4-5	-16.013	-977.634	961.621						
21				5-1	-382.179	-382.179	0.000						
22													
23				SUM	665.340	0.000	665.340						
24													
25				ENGINE SIZING						Turbo Activated ?		no	
26	Volumentic eff	1.100			4.000	No. Cycles			TURBOCHARGER STUFF				
27	2 stroke or 4 stroke engine?	4.000			3.150	Bore			Tmin Exh. Cooling [R]				
28	Engine speed(rpm)	15000.000			2.087	Stroke			Ambient Temp				
29	Engine disp. (in^3)	97.600				No. of Engines	1.000		Ambient Pressure				
30	Engine power(hp, kW)	1027.045	765.880			Radiator DT	25.000		Cycle cooling coeff.				
31	Engine torque(Ft-Lbf)	359.603				Twater,in	175.000		Turbo Expander Efficiency				
32	Brake Mean Eff Pressure (psia)					Twater,out			Turbo Bypass "f"				
33	Piston speed(ft/min,m/s)					PUMP EFF	0.850		Compressor Efficiency				
34	Heat content(btu/ft^3)	899158.000				Rad. Ht Rec. Massflow			Cycle Mix Temp (T1) R				
35	Air/Fuel ratio	14.000				Dp=cond	10.000		delta H				
36	Mair	3886.44	LBm/Hr			hp	750.000		Texhaust Out (R, F)				
37	Mfuel	220.46	LBm/Hr						Tmix (R, F)				
38						Closed Mass Energy							
39						Seconds/Cycle			Pout (Psia)				
40						BSFC (LBm/Hr/Hp)			@Tout (R, F)				
									99.14074566				
									1011.36184				
									551.3618398				

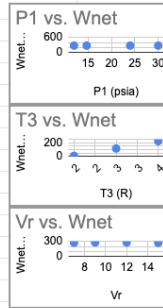


Figure 7: Regular Engine Operation WITH Turbocharger Activated

	A	B	C	D	E	F	G	H	I	J	K	L
1	Vr	12.000		NOTE: ANYTHING IN BOLD IS INPUT. ALL ELSE DEPENDS ON INPUT IF k == comprss index or expansionindex, there is no heat transfer Internal energy = Cv(deltaT)					Vr compression ratio v1/v2			
2	Vc	1.000							Vc cutoff ratio v4/v3			
3	k	1.412							Cp specific hear constant p			
4	Cp	0.240							Cv specific heat constant v			
5												
6	Cv	0.170		PROPERTY TABLE					k specific heat ratio Cp/Cv			
7	Mole Wt.	28.966			p(psia)	T(R)	v(ft^3/Lbm)					
8	Ru	1545.000		1.000	99.000	1011.000	3.783		(n-1)/n, expansion	0.2907801418		
9	Compress. Index	1.410	n, 1-2 nc	2.000	3290.643	4279.498	0.315					
10	Expansion Index	1.410	n, 4-5 ne	3.000	10575.668	9000.000	0.315					
11	otto Cycle Eff.	0.474		4.000	10575.668	9000.000	0.315					
12	Cycle Efficiency	0.626		5.000	318.172	3249.213	3.783					
13	Efficiency	0.825										
14	Eng. Eff Corr		<--WorkOUT/HeatIN									
15	Carnot Effic	0.888		PROCESS TABLE								
16	Exergy in Exhaust				Heat	Internal Energy	Work					
17	Eff w losses	0.450		1-2	256.434	555.645	-299.210					
18	BSFC(LBm/hr/hp)			2-3	802.485	802.485	0.000					
19	Intercooler effectiveness			3-4	0.000	0.000	0.000					
20	Closed mass energy			4-5	-16.013	-977.634	961.621					
21				5-1	-380.496	-380.496	0.000					
22												
23				SUM	662.410	0.000	662.410					
24												
25				ENGINE SIZING					Turbo Activated ? no			
26	Volumentic eff	1.100			4.000	No. Cycles			TURBOCHARGER STUFF			
27	2 stroke or 4 stroke engine?	4.000			3.150	Bore			Tmin Exh. Cooling [R]			
28	Engine speed(rpm)	15000.000			2.087	Stroke			Ambient Temp 540			
29	Engine disp. (in^3)	97.600			No. of Engines	1.000			Ambient Pressure 14.7			
30	Engine power(hp, kW)	1048.769	782.080		Radiator DT	25.000			Cycle cooling coeff. 0.82			
31	Engine torque(Ft-Lbf)	367.209			Twater,in	175.000			Turbo Expander Efficiency 0.85			
32	Brake Mean Eff Pressure (psia)				Twater,out				Turbo Bypass "F" 0.71			
33	Piston speed(ft/min,m/s)				PUMP EFF	0.850			Compressor Efficiency 0.85			
34	Heat content(btu/ft^3)	899158.000			Rad. Ht Rec. Massflow				Cycle Mix Temp (T1) R 676.6677602			
35	Air/Fuel ratio	14.000			Dp=cond	10.000			delta H 391.7475494			
36	Mair	3086.44	LBm/Hr		hp	750.000			Texhaust Out (R, F) 1616.932 1156.932			
37	Mfuel	220.46	LBm/Hr						Tmix (R, F) 2775.851501 2315.851501			
38					Closed Mass Energy							
39					Seconds/Cycle				Pout (Psia) 99.75821356			
40					BSFC (LBm/Hr/Hp)				@Tout (R, F) 1013.361622 553.361622			

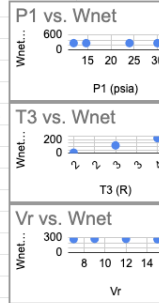


Figure 8: Regular Engine Operation WITH Turbocharger Activated

	A	B	C	D	E	F	G	H	I	J	K	L
1	Vr	12.000		NOTE: ANYTHING IN BOLD IS INPUT. ALL ELSE DEPENDS ON INPUT IF k == comprss index or expansionindex, there is no heat transfer Internal energy = Cv(deltaT)					Vr compression ratio v1/v2			
2	Vc	1.000							Vc cutoff ratio v4/v3			
3	k	1.412							Cp specific hear constant p			
4	Cp	0.240							Cv specific heat constant v			
5												
6	Cv	0.170		PROPERTY TABLE					k specific heat ratio Cp/Cv			
7	Mole Wt.	28.966			p(psia)	T(R)	v(ft^3/Lbm)					
8	Ru	1545.000		1.000	100.000	1013.400	3.754		(n-1)/n, expansion	0.2907801418		
9	Compress. Index	1.410	n, 1-2 nc	2.000	3323.882	4279.498	0.313					
10	Expansion Index	1.410	n, 4-5 ne	3.000	10657.194	9000.000	0.313					
11	otto Cycle Eff.	0.474		4.000	10657.194	9000.000	0.313					
12	Cycle Efficiency	0.626		5.000	320.625	3249.213	3.754					
13	Efficiency	0.825										
14	Eng. Eff Corr		<--WorkOUT/HeatIN									
15	Carnot Effic	0.887		PROCESS TABLE								
16	Exergy in Exhaust				Heat	Internal Energy	Work					
17	Eff w losses	0.450		1-2	255.316	555.237	-299.921					
18	BSFC(LBm/hr/hp)			2-3	802.485	802.485	0.000					
19	Intercooler effectiveness			3-4	0.000	0.000	0.000					
20	Closed mass energy			4-5	-16.013	-977.634	961.621					
21				5-1	-380.088	-380.088	0.000					
22												
23				SUM	661.700	0.000	661.700					
24												
25				ENGINE SIZING								
26	Volumentic eff	1.100				4.000	No. Cycles		Turbo Activated ?	no		
27	2 stroke or 4 stroke engine?	4.000				3.150	Bore		TURBOCHARGER STUFF			
28	Engine speed(rpm)	15000.000				2.087	Stroke		Tmin Exh. Cooling [R]			
29	Engine disp. (in^3)	97.600				No. of Engines	1.000		Ambient Temp	540		
30	Engine power(hp, kW)	1056.853	788.108			Radiator DT	25.000		Ambient Pressure	14.7		
31	Engine torque(Ft-Lbf)	370.040				Twater,in	175.000		Cycle cooling coeff.	0.82		
32	Brake Mean Eff Pressure (psia)					Twater,out			Turbo Expander Efficiency	0.85		
33	Piston speed(ft/min,m/s)					PUMP EFF	0.850		Turbo Bypass "f"	0.71		
34	Heat content(btu/ft^3)	899158.000				Rad. Ht Rec. Massflow			Compressor Efficiency	0.85		
35	Air/Fuel ratio	14.000				Dp=cond	10.000		Cycle Mix Temp (T1) R	676.6677602		
36	Mair	3886.44	LBm/Hr			hp	750.000		delta H	392.352216		
37	Mfuel	220.46	LBm/Hr						Texhaust Out (R, F)	1614.412	1154.412	
38									Tmix (R, F)	2775.120862	2315.120862	
39						Closed Mass Energy			Pout (Psia)	99.98448982		
40						Seconds/Cycle			@Tout (R, F)	1014.092261	554.092261	
						BSFC (LBm/Hr/Hp)						

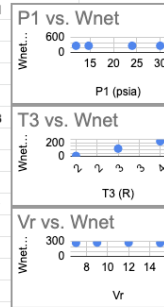


Figure 9: Regular Engine Operation WITH Turbocharger Activated

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